Capstone Project

Build and Present a Full
MLOps Solution on Azure

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Introduc

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Beluga Journey

Pipeline Operations

Content Map

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TDC: AI **Level:** Practitioner **User:** Implementer **Persona:** MLOps **Theory / Practice: 40/60**

Duration: 5-10 weeks | No. of training days: 5 | No. of modules: 4

Focus points:

- Lecture / Knowledge
- Key Exercise / Skill
- Inspiration / Attitude

DAY 1

1 Beluga Introduction ML & MLOps Essentials

Why is managing the full lifecycle of AI models critical in enterprise environments?

2 Hands-on ML Workshop

What does a successful ML model development process look like in practice?

DAY 2

3 Experiment Tracking

What are best practices for managing experiments and model versions?

4 MLOps Deep Dive

How do CI/CD/CT practices apply to machine learning pipelines?

5 Cloud Infrastructure for MLOps I & II

How can we balance control, flexibility, and cost in our MLOps infrastructure?

DAY 3

6 Deploying and Managing AI Models in the Cloud

How to use all the Digital Competencies to your benefit?

DAY 4

7 Legal and Ethical Frameworks for AI

How to build AI systems that are both ethical and auditable?

8 Introduction to Responsible AI

How do we build fairness and privacy into the AI lifecycle?

DAY 5

9 Capstone Project

How to design and implement an end-to-end AI pipeline that meets OTP's needs?

10 Closure and Takeaways

How can we apply what we've learned to accelerate real project delivery?

Agenda

- Introduction, task explanation, team formation
- Brainstorming & design
- System Architecture Setup
- Data analysis and model development
- Model deployment & monitoring setup
- Model maintanance
- Finalize solution, prepare presentation
- Team presentations & feedback

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MLOps Tasks – Overview

Each team will build a **production-ready ML system** on Azure, covering

- **Data ingestion** & preprocessing
- **Model training** (initial + retraining support)
- **Model versioning** & evaluation
- **Deployment** (real-time or batch)
- **Monitoring** (drift, latency, errors, etc.)
- **Re-training**
- **Cost, scalability, and ethical review**

Team Structure and Rules

- Teams of **3–4 people**
- Deliverables:
  - Working **MLOps pipeline** on Azure
  - Business Requirements Document
  - Responsible AI Checklist
  - High-Level Architecture Document
  - A **5-minute presentation** of your project
- You can choose one of the following projects:
  - Credit Default Prediction
  - Medical Insurance Cost Prediction

Use Case 1 - Credit Default Prediction

Objective

- Predict which clients are at high risk of defaulting on their payment next month.
- Enable proactive risk management and collection strategies.

- **Problem Type**: Binary Classification (Default / Not Default)
- **Dataset**: Client demographics, credit limits, and 6-month payment/bill history.

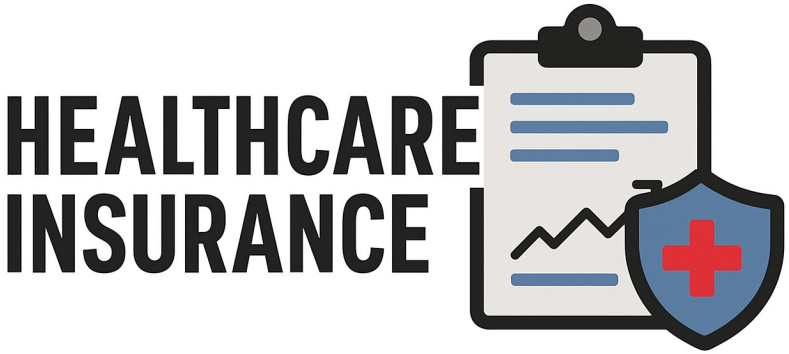
Use Case 2 - Healthcare Insurance

Objective

- Predict a new customer's potential medical insurance costs.
- Inform premium pricing and risk underwriting decisions.

Problem Type: Regression

Dataset: Personal attributes (age, sex, BMI, no. children, smoker) and geographic region.



Evaluation

Your project will be evaluated based on the following criteria:

- **End-to-End Functionality**
  - A working MLOps pipeline from data to deployment and monitoring
- **Code Quality**
  - Clean, modular, reusable code with good structure and documentation
- **Monitoring & Retraining**
  - Evidence of logging, model monitoring, and support for retraining
- **Ethical & Legal Considerations**
  - Discussion of fairness, bias, and regulatory aspects
- **Presentation & Communication**
  - Clear explanation of the approach, decisions, and outcomes

Team Presentations & Feedback

Each team will deliver a short presentation including:

🧩 **System Architecture**

- Visual overview of your MLOps pipeline

⚙️ **MLOps Implementation**

- Tools used, automation, deployment, and monitoring setup

🚧 **Key Challenges & Learnings**

- What worked, what didn't, and how you adapted

💻 **Live Demo (if possible)**

- Walkthrough of key components or running system

📣 **Feedback Session**

- Constructive feedback from instructors and peers

🎉 **Celebrate Your Success!**

- Acknowledge the hard work and teamwork